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A reinforcement learning model for dynamic distribution route optimization of agricultural products under replenishment demand

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ABSTRACT

Most existing logistics optimization models assume static demand or ignore transportation replenishment, which limits their applicability in dynamic agricultural environments. Therefore, this paper constructs a dual-objective multi-center cold chain distribution model to minimize total cost and maximize customer satisfaction. It incorporates dynamic customers and in-transit replenishment into the constraints and introduces a Reinforcement Learning-guided Adaptive Large Neighborhood Search (RL-ALNS) algorithm for the solution. Actor-Critic deep reinforcement learning designs state-action spaces and reward functions for dynamic delivery environments. It iteratively updates paths by jointly training strategy networks and value networks. Then, it embeds reinforcement learning-guided adaptive large neighborhood search into the solution process, dynamically selecting destruction-repair operators to improve convergence efficiency and stability in large-scale, real-time updating scenarios. Experimental results demonstrate that, under the same customer scale, allowing vehicles to restock en-route can reduce total costs from 2920.12 RMB to 2246.32 RMB, while improving customer satisfaction from 0.8613 to 0.9346. For a medium-sized instance with 52 nodes, the optimal cost achieved after five independent runs of the algorithm is 2246.64 RMB. In small-scale benchmark tests, the RL-ALNS reduces the solution time from 28 s to 7.04 s.

1. Introduction

With the rapid development of China's rural economy, the agricultural product logistics industry [1] has gained significant momentum. However, persistent challenges remain in dynamic logistics distribution, particularly in reducing exorbitant transportation costs [2] and improving operational efficiency. The accelerating pace of modern life has increased the frequency of dynamic order changes, making adaptive distribution strategies [3] a critical breakthrough for industry advancement. Establishing a dynamic route optimization method [4] that incorporates replenishment demands has emerged as a pivotal approach to enhance corporate competitiveness [5].

Recent studies highlight the growing attention towards multi-distribution center logistics models with dynamic demand and en-route replenishment. Goel et al. [6] considered the probabilistic elasticity of service orders and service times, and developed a mathematical model aimed at minimizing transportation costs while maximizing

customer satisfaction. Talouki et al. [7] investigated the dynamic green vehicle routing problem for perishable products under green transportation conditions, optimizing the total cost of dynamic traffic networks, effectively reducing environmental impacts, and improving customer satisfaction. Bertazzi et al. [8] studied the vehicle routing problem with replenishment. When the remaining vehicle load is insufficient to support the subsequent service plan, vehicles are allowed to return to the depot for replenishment. Follow-up tests showed that adopting replenishment strategies in distribution schemes can improve overall transportation efficiency.

Machine learning algorithms have revolutionized Vehicle Routing Problem (VRP) solutions [9]. In the research on multi-distribution center logistics problems, Lahyani et al. [10] applied an adaptive large neighborhood search algorithm to solve the open multi-depot problem. They tested the approach on multiple datasets of different scales, with the largest case involving six depots and 288 customer points, and demonstrated the adaptability and superiority of the algorithm in

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solving such problems. Husakou et al. [11] studied the open routing problem of heterogeneous multi-depot fleets, where different fleets have varying unit transportation costs and load capacities. Their model aimed at minimizing transportation costs, and practical instances were solved to optimality using a standard mixed-integer programming solver. Sadati et al. [12] investigated the multi-depot green routing optimization problem and developed a mathematical model that minimizes the total transportation distance. They designed a hybrid algorithm combining tabu search and generalized variable neighborhood search, and computational experiments on small-scale cases proved the strong robustness of the proposed algorithm.

Through reasonable analysis of the problem, this paper constructs a more efficient distribution path model that takes into account the changes of customer demand, multi-distribution center distribution, and replenishment strategy along the way, designs suitable algorithm, completes reasonable organization and plan, and has certain research innovation.

- (1) This paper combines the Actor-Critic deep reinforcement learning model with the ALNS framework to propose a hybrid solution framework called the Reinforcement Learning guided-Adaptive Large Neighborhood Search (RL-ALNS). It uses reinforcement learning to dynamically select destruction and repair operators, overcoming the problem of traditional ALNS operator selection relying on human experience.
- (2) A reward function and weight allocation mechanism for multi-objective optimization have been designed. The reward function comprehensively considers transportation costs, time window penalties, energy consumption, and agricultural product quality loss, transforming the multi-objective optimization problem into a form that can be handled by reinforcement learning. At the same time, a weight allocation method has been introduced to flexibly adjust the trade-off between cost minimization, customer satisfaction maximization, and energy consumption reduction under different operating conditions.
- (3) A dynamic route optimization training process based on rolling time domain updates has been designed. Dynamic demand is divided into multiple time windows, and reinforcement learning is combined with online route plan updates to enable real-time response to dynamic customer demand and in-transit replenishment.

In this paper, the multi-distribution model of agricultural products logistics distribution center considering dynamic demand and replenishment along the way is constructed and the corresponding search algorithm is designed to solve them effectively, aiming to improve the service ability of agricultural products distribution enterprises and improve customer satisfaction. It has the great practical significance and the application value of maintaining the core competitiveness of logistics enterprises.

2. Related work

2.1. Cold chain logistics distribution models

In the early 20th century, refrigeration technology was invented, which marked the beginning of modern cold chain logistics. This breakthrough enabled large-scale transportation and brought considerable economic benefits. Up to now, the development of cold chain logistics abroad has become relatively mature. Scholars have carried out extensive research focusing on distribution optimization and distribution technologies in cold chain logistics, achieving fruitful results.

Likar et al. [13] measured the temperature conditions of 1688 perishable food cooling devices, confirming that poor temperature control during cold chain distribution is an important cause of food spoilage. They further emphasized that customer satisfaction with

product quality has a significant impact on cold chain retailers. Accorsi et al. [14] proposed a mixed-integer linear programming model that incorporates meteorological conditions into the planning of production, storage, and distribution of perishable products, given the high sensitivity of cold chains to ambient temperature. Case studies verified that with rising ambient temperatures, the model can significantly reduce both energy consumption and costs. Mejjaoui et al. [15] developed a real-time responsive cold chain distribution scheduling decision model along with an RFID-based network monitoring system. This system can promptly adjust distribution time and routes when emergencies occur during the routine transportation of perishable goods. Case studies demonstrated that it effectively reduces costs. Hsu et al. [16] considered different delivery time windows in cold chain transportation and constructed a multi-temperature joint distribution model. The model determined the optimal delivery cycle for foods in different temperature ranges, providing valuable reference for enterprises to improve service levels and distribution efficiency. Stellingwerf et al. [17] improved the traditional vehicle routing problem model that accounts for load by additionally considering the impact of refrigeration emissions on vehicle speed and total carbon emissions. Validation results showed that the improved model can effectively reduce fuel consumption and overall costs. Reza et al. [7] investigated the dynamic green vehicle routing problem for perishable products under green transportation conditions. Their model optimized the total cost of dynamic traffic networks, effectively reduced environmental impact, and enhanced customer satisfaction.

In terms of algorithmic approaches, Li et al. [18] considered the dynamic variations in customer demand and road conditions and designed a heuristic algorithm based on tabu search. Experiments demonstrated that the algorithm can effectively solve such stochastic vehicle routing problems. Brito et al. [19] addressed the vulnerability of cold chain interruptions by employing a fuzzy method combined with a hybrid heuristic approach to solve flexible cold chain distribution problems with uncertain time constraints. The proposed method achieved promising validation results on small-scale examples. Moreover, low-carbon and sustainability considerations have become an important research direction in cold chain logistics. Liu et al. [20] established an optimization model for joint distribution vehicle routing under a carbon trading mechanism. Results showed that compared with single distribution, joint distribution can significantly reduce total costs and carbon emissions. Ning et al. [21] studied the optimization of fresh agricultural product cold chain distribution routes under a carbon tax mechanism. They proposed an improved quantum bacterial foraging optimization algorithm, using carbon tax cost as a decision variable. Simulation and comparative experiments confirmed that the improved algorithm effectively optimized distribution routes while reducing both carbon tax costs and overall costs.

2.2. Fundamental extensions of the vehicle routing problem

Dantzig and Ramser [22] first introduced the Vehicle Routing Problem, expanding the research scope from a single-route model to a multi-route framework, which significantly improved the practical applicability of distribution planning. After decades of development, the theoretical foundations of VRP have become relatively mature, and scholars have increasingly focused on extending VRP to reflect realistic operational constraints.

One important line of research incorporates time window constraints into VRP. Azi et al. [23] emphasized the role of time windows in distribution problems, showing that the introduction of hard time windows increases the number of customers that can be served. Tas et al. [24] proposed the use of soft time windows, which, although slightly increasing costs, greatly enhanced customer satisfaction. Chao et al. [25] combined time windows with inventory decisions and developed a two-stage location-routing problem where soft time windows were used to measure the penalty cost of deviations in delivery time. Duan et al.

[26] further considered variations in travel times, constructing a multi-objective model with time windows and demonstrating its robustness through a particle swarm optimization algorithm.

Research has also focused on VRP under dynamic environments. AbdAllah et al. [27] applied a time-segmentation approach to handle dynamic path optimization and designed an improved genetic algorithm. Çimen et al. [28] developed a mathematical model that incorporates both vehicle speed variation and carbon emissions, and proposed a heuristic algorithm to solve it, showing that the method is suitable for problems of different sizes and has strong applicability in sustainable logistics decision-making. Goel et al. [29] introduced the probability elasticity of service orders and service times into a mathematical model aimed at minimizing transportation costs and maximizing customer satisfaction, effectively addressing failures caused by capacity and time window limitations. Oliveira et al. [30] studied dynamic VRPs involving traffic congestion and varying order quantities, comparing the performance of the ant colony algorithm in static and dynamic settings, and concluded that appropriate parameter tuning combined with local search significantly improves its adaptability in dynamic contexts.

2.3. Green logistics and replenishment strategies in VRP

In parallel with these foundational extensions, the consideration of environmental sustainability and operational replenishment strategies has become a growing focus in VRP research. A key stream of work concerns carbon emissions in VRP. Saif et al. [31] proposed a mixed-integer linear programming model for the cold chain industry, aiming to minimize both costs and carbon emissions, and validated its effectiveness through simulation experiments. Musavi et al. [32] examined a multi-objective food supply chain scheduling problem, calculating emissions separately for empty and fully loaded vehicle operations. Xu et al. [33] estimated emissions based on fuel consumption and developed a mixed-integer nonlinear programming model that balances fuel efficiency with customer satisfaction, proving that fuel use can be substantially reduced without compromising service quality.

Another important dimension is the study of multi-depot VRPs. Shen et al. [34] considered open multi-depot routing under carbon trading policies and time windows, proposing a model that minimizes total costs, including carbon emissions and driver wages. A two-stage algorithm was designed and sensitivity analysis revealed that carbon taxes and prices significantly influence total distribution costs. Lahyani et al. [10] employed an adaptive large neighborhood search algorithm to solve open multi-depot problems, and extensive computational testing confirmed its strong performance. Husakou et al. [11] investigated heterogeneous multi-depot fleets with varying capacities and unit transportation costs, solving practical cases to optimality using a standard mixed-integer programming solver. Sadati et al. [12] focused on multi-depot green routing optimization, developing a hybrid tabu search and variable neighborhood search algorithm that proved to be highly robust in small-scale tests.

Finally, replenishment strategies have been incorporated into VRP research. Tarantilis et al. [35] examined routing with intermediate replenishment facilities and proposed an improved tabu search algorithm integrated with variable neighborhood search, which performed well across multiple test datasets. Bertazzi et al. [8] addressed stochastic demand with replenishment, showing that allowing vehicles to return to depots for restocking significantly improves overall efficiency. Wang et al. [36] studied a multi-trip, multi-compartment replenishment VRP in which fuel trucks distribute fuel to gas stations and may return to the depot when capacity is insufficient. A model was established with the objective of minimizing total travel distance, and the proposed algorithm demonstrated strong stability in computational experiments.

3. Methodology for dynamic distribution route optimization

3.1. Determination of the objective function

The problems studied in this paper can be described as follows. Within a specified time period, a certain number of customers are served jointly by multiple distribution centers in a given area. The distributed products are agricultural goods, and the objective is to satisfy demand through reasonable planning of vehicle distribution routes. The customers are divided into two parts, the one is static customers and the other is dynamic customers. Static customers refer to those who place orders the day before delivery begins, while dynamic customers are new orders that appear in real time after delivery has started. Because dynamic demand may lead to vehicle capacity shortages or stock-outs at the original distribution center, vehicles are allowed to replenish at any distribution center in the area. This paper considers a multi-center distribution model with dynamic demand and replenishment en-route, and formulates it as a dual-objective reinforcement learning adaptive large neighborhood search (RL-ALNS) algorithm. The objectives are to minimize the total cost while maximizing customer satisfaction.

The transportation of agricultural products is characterized by large quantity and many loading and unloading times. Multiple factors such as quality loss rate, total transportation cost, transportation energy consumption, and dynamic distribution loss should also be considered in the distribution path selection model [37] of agricultural products. The quality loss rate is closely related to the transportation time, and the prolonged transportation time will lead to the decline of the quality of agricultural products. The transportation time is also closely related to transportation efficiency and customer satisfaction. The energy consumption of transportation means determines the cost of agricultural products transportation.

Qualitative damage rate. The damage rate of agricultural products in the process of transportation mainly includes node replacement damage and transportation damage. Generally, the quality integrity of agricultural products meets the following relationship of t in Eq. (1).

$$\lambda = \min \left\{ \frac{1 - \frac{t}{t_B}}{1 - QP}, 1 \right\} \quad (1)$$

where QP is the quality decline rate of agricultural products, t_B is the shelf life of agricultural products, t is the total transportation time. This paper assumes that the quality of all agricultural products is stable, and the environment of agricultural products is stable during transportation.

Based on the above analysis, ratio W_1 can be expressed as shown in Eq. (2).

$$W_1 = \varphi \left\{ 1 - \frac{1 - \frac{t}{t_B}}{1 - QP} \right\} \quad (2)$$

where φ is the quality loss discount coefficient for agricultural products.

Total cost of transportation. The total transportation cost in the process of agricultural products distribution includes vehicle transportation fee, replacement fee, and time punishment fee. In this paper, when choosing the path of agricultural products distribution network without considering the transportation price difference between different regions, then the transportation cost can be expressed as shown in Eq. (3).

$$C_1 = \sum_{i,j \in M} \sum_{k \in n} w^k \cdot d_{ij}^k \cdot q \cdot x_{ij}^k \quad (3)$$

where M is the set of transport nodes, i, j is the transport node, n is the number of transport nodes, w^k is the transport cost under the transport mode in k , d_{ij}^k is the transport distance under the transport mode in k , q is

the quantity of goods (box), x_{ij}^k is the decision variable (take 0 or 1).

Change fee is to point to agricultural products in transportation, repair, loading and unloading, stacking, and other operations of the cost. These costs often according to the length of goods transportation and the characteristics of different types of goods are different. According to the needs of different goods, the cost expression as Eq. (4).

$$C_2 = \sum_{j \in M} \sum_{k, l \in n} w^{kl} \cdot q \cdot y_j^{kl} \quad (4)$$

where w^{kl} is the replacement cost between different transport modes and y_j^{kl} is the decision variable (take 0 or 1).

Time penalty cost refers to the increase of the transportation cost of agricultural products caused by the transportation time of agricultural products. Usually, the time penalty cost of agricultural products can be expressed as Eq. (5).

$$C_3 = q \cdot D \cdot \max(\alpha_1 - t, 0) + q \cdot G \cdot \max(t - \alpha_2, 0) \quad (5)$$

where D is the penalty cost for early arrival of agricultural products, G is the penalty cost for delayed arrival of agricultural products, $[\alpha_1, \alpha_2]$ is the satisfactory arrival time interval of the customer

Through the above analysis, the total transportation cost of agricultural products is shown in Eq. (6).

$$W_2 = C_1 + C_2 + C_3 \quad (6)$$

Total transportation energy consumption. During the entire distribution and transportation process of agricultural products, both transportation energy consumption and equipment energy consumption are generated. Transportation energy consumption includes the energy used for vehicle operation as well as the energy consumed at transit nodes.

Agricultural products in the transit node to maintain fresh agricultural products energy consumption is shown in Eq. (7).

$$E_1 = \frac{H \cdot S_e}{1000} \cdot \sum_{i \in M} \sum_{k \in n} \frac{d_{ij}^k}{v^k} \cdot x_{ij}^k \quad (7)$$

where H is the preservation coefficient of agricultural products, S_e is the transportation fuel consumption rate (g / kw•h), and v^k is the average transportation cost under the transportation mode in k .

The energy consumption generated when the agricultural products are placed or replaced at the transportation node can be expressed as Eq. (8).

$$E_2 = \sum_{i \in M} \sum_{k \in n} (t^{kl} + tw^{kl}) \cdot y_j^{kl} \quad (8)$$

where t^{kl} is the total time (h) of agricultural products at the transit node, w^{kl} is the cost of converting the transportation mode of agricultural products at the transit node.

The equipment loss is generated when the agricultural products are stored, transported, or replaced at the transfer node. And the calculation formula is Eq. (9).

$$E_3 = \sum_{i \in M} \sum_{k \in n} d_{ij}^k \cdot c^k \cdot x_{ij}^k \cdot p \quad (9)$$

where c^k is the energy consumption of the transport mode in k (kg / t•km).

Dynamic distribution loss. In the process of transportation of agricultural products, there may be a large number of dynamic customers,

which will lead to the shortage of transport vehicles or the lack of stock in the distribution center. At this time, the vehicle needs to go to another distribution center to replenish the goods, resulting in excess distribution loss. This paper abstracts the practical problem into a cold chain logistics distribution model that considers both dynamic demand and en-route replenishment at distribution centers. Since the problem is complex and requires the joint consideration of dynamic demand, multi-center distribution, and vehicle replenishment, simplifications are introduced to ensure tractability and operability. In this context, dynamic distribution losses can be defined as shown in Eq. (10).

$$E_4 = \sum_{i \in M} \sum_{k \in n} f_{ij}^k \cdot mQ \quad (10)$$

where f_{ij}^k is the energy consumption coefficient of dynamic transport demand in the transport mode in k , m is the quantity of dynamic transport demand, Q is the energy consumption of a single dynamic demand distribution

Therefore, the total energy consumption of agricultural products transportation considered in this paper can be expressed as Eq. (11).

$$W_3 = E_1 + E_2 + E_3 + E_4 \quad (11)$$

Based on the automatic selection model of agricultural products, the above analysis identifies the key influencing factors in agricultural product transportation.

3.2. Way selection algorithm design

In optimizing the optimization of dynamic distribution path of agricultural products, this paper comprehensively considers the time window problem, customer location information, and express node information under dynamic replenishment demand. These influencing factors are quantified, and a calculation model with an enhanced framework is designed. In this approach, training the policy network and value network models, only by observing the reward signal and following the feasibility rule, finds near-optimal solutions [38] for problem samples sampled from a given distribution. In summary, combined with the objective function, the following model can be established as Eqs. (12)-(14).

$$\min W_1 = \varphi \left(1 - \frac{1 - \frac{t}{t_b}}{1 - QP} \right) \quad (12)$$

$$\min W_2 = \min(C_1 + C_2 + C_3) \quad (13)$$

$$\min W_2 = \min(E_1 + E_2 + E_3) \quad (14)$$

The constraints of the model are as follows.

In the process of agricultural products transportation, a mode of goods transportation is adopted for node replenishment transportation as Eq. (15).

$$\sum_{k \in n} x_{ij}^k \leq 1 \quad (15)$$

In the node replenishment transportation, the number of replacement operations of agricultural products should not exceed once as Eq. (16).

$$\sum_{k \in n} \sum_{l \in n} y_j^{kl} \leq 1 \quad (16)$$

In the transportation of agricultural products, the complete transportation path is conducted from point O to point D, which is shown in Eq. (17).

$$\sum_{k \in n} \sum_{i \in Q} x_{ij}^k = \begin{cases} -1, \forall j \in O \\ 0, \forall i \in Q \\ 1, \forall j \in D \end{cases} \quad (17)$$

Hard time window constraints for the end of produce like $t_{\min} \leq t \leq t_{\max}$.

In the whole transportation process, it is necessary to ensure that replenishment and the overall transportation network maintain continuity as shown in Eq. 18.

$$\sum_{i \in Q} x_{ij}^k \bullet d_{ij}^k = y_j^{kl} \quad (18)$$

In summary, this paper develops an optimization algorithm for the dynamic distribution path of agricultural products considering replenishment demand. To obtain the optimal solution, the multi-objective model is transformed into a single-objective model. The transformation method adopted in this paper is the weight allocation method, which is shown in Eq. (19).

$$\min W = \min \alpha W_1 + \min \beta W_2 + \min \gamma W_3 \quad (19)$$

where α , β , and γ are the weights of models W_1 , W_2 , W_3 , and $\alpha + \beta + \gamma = 1$.

The weight allocation method (α , β , γ) introduces trade-offs between cost minimization, satisfaction maximization, and energy efficiency. For instance, higher β prioritizes cost reduction but may lower satisfaction. Future work will explore Pareto-optimal solutions to quantify these trade-offs systematically.

3.3. Deep reinforcement learning training process based on the Actor-Critic architecture

Through the above design of the dynamic distribution path optimization algorithm of agricultural products considering the capture requirements, it is not difficult to find that it is difficult to solve the algorithm with the traditional linear solution method, so the deep reinforcement learning algorithm is introduced in this paper. The algorithm adopts deep reinforcement learning (DRL) [39], which integrates neural networks into the reinforcement learning framework to improve decision-making efficiency and perception of dynamic environments in the original algorithm, so as to help the model to obtain the optimal solution.

To address the path optimization problem for agricultural products in a dynamic delivery environment, this paper adopts the Actor-Critic (AC) approach and combines it with a round-based update deep reinforcement learning algorithm for network training. The random policy network is modeled using parameters θ , while the Critic network is parameterized using φ to estimate the expected return value of the policy execution. The gradient of the policy parameters with respect to the estimated predicted return is used to iteratively improve the policy. The core idea is to construct an objective function based on a predefined reward function, aiming to reduce total transportation costs, losses, and time penalties to enhance the timeliness and quality stability of agricultural products during the delivery process. This enables the joint training of the policy network and value network. The training process of this model relies on the state-action distribution within the reinforcement learning framework. In each training iteration, θ represents all trainable variables in the model, corresponding to the random

delivery paths generated based on the current parameters.

First, randomly generate a batch of training instances with time windows, each instance containing a fixed distribution center and static/dynamic customer demand information. The batch size is M , and the batch represents the data scale relied upon for model parameter updates. During training, initialize the policy network, i.e., the Actor network and Critic network, and use the policy to obtain a batch of instance solutions, thereby obtaining the reward. The reward is composed of multiple factors, including transportation costs, time window violation penalties, and quality degradation of agricultural products. The Critic network is used to estimate the expected return from the current policy, which is used to construct the gradient of the objective function.

The policy network is updated through backpropagation of these reward signals, while the critic network is iteratively optimized by minimizing the return estimation error. The addition of the critic improves training stability and convergence speed, especially in complex state transition processes in multi-objective optimization scenarios for agricultural products, where it can more effectively guide the direction of policy updates.

The objective function is defined as maximizing expected reward, which essentially corresponds to minimizing total loss under multi-objective optimization, including transportation costs of delivery routes, vehicle dispatch frequency, time window violation losses, and quality degradation levels. In the model, it is set as the negative expected total return of the trajectory Y sampled by the random strategy, as shown in Eq. 20.

$$J(\theta) = E_{\pi^\theta} [R(Y)] \quad (20)$$

where $R(Y) = -(\alpha W_1 + \beta W_2 + \gamma W_3)$.

The loss function for the gradient of the policy network parameters for M delivery problem instances using Monte Carlo sampling is shown in Eqs. 21 and 22.

$$\nabla_\theta J(\theta) = \frac{1}{M} \sum_{i=1}^M [R(Y_{[i]}) - v_\varphi(X_{[i]})] \nabla \log P_\theta(Y_{[i]} | X_{[i]}) \quad (21)$$

$$P_\theta(Y_{[i]} | X_{[i]}) = P_\theta(y_{[i]}^1 | X_{[i]}^0, G_{[i]}^0, y_{[i]}^0) \cdots P_\theta(y_{[i]}^{Y_{[i]}} | X_{[i]}^{Y_{[i]}-1}, G_{[i]}^{Y_{[i]}-1}, y_{[i]}^{Y_{[i]}-1}) \quad (22)$$

According to the probability chain derivation formula and the free model, we obtain $P_\theta(y_{[i]}^{t+1} | X_{[i]}^t, G_{[i]}^t, y_{[i]}^t)$. Where M is the batch size. θ is the policy network parameter. X is the i -th instance in the batch. R is the expected return of the actual trajectory of the policy in the i -th instance. v_φ is the Critic network that approximates $R(Y_{[i]})$. Next, we use the Adam optimizer to update the parameters by gradient ascent according to Eq. 23 and update the parameters by gradient descent according to Eq. 25.

$$\theta = \theta + \alpha_1 \frac{1}{M} \sum_{i=1}^M [R(Y_{[i]}) - v_\varphi(X_{[i]})] \nabla \log P_\theta(Y_{[i]} | X_{[i]}) \quad (23)$$

$$d\varphi = \frac{1}{M} \sum_{i=1}^M \nabla_\varphi [R(Y_{[i]}) - v_\varphi(X_{[i]})]^2 \quad (24)$$

$$\varphi = \varphi + \alpha_2 \frac{1}{M} \sum_{i=1}^M \nabla_\varphi [R(Y_{[i]}) - v_\varphi(X_{[i]})]^2 \quad (25)$$

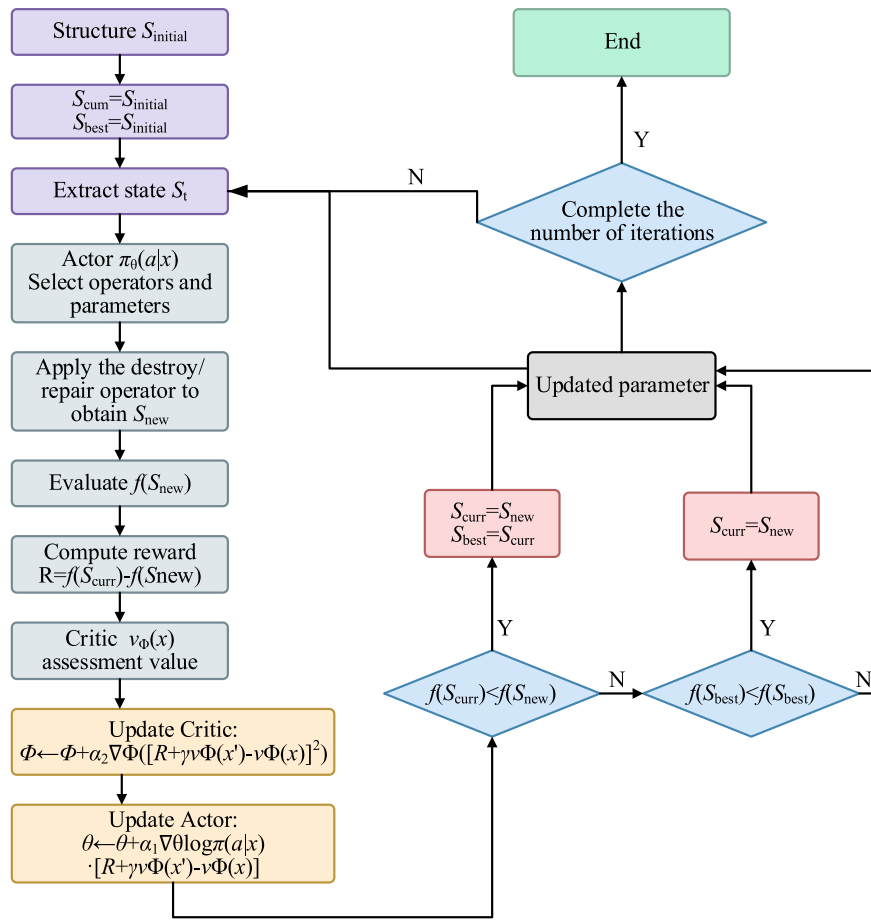


Fig. 1. Flow chart of the RL-ALNS algorithm.

Table 1

The pseudocode of RL-ALNS algorithm.

RL-ALNS
Input: $S_{initial}$, $D = d_1, \dots, d_k$, $R = r_1, \dots, r_h$, $f(\cdot)$, T , Actor $\pi_\theta(a s)$, Critic $v_\phi(s)$ Output: S_{best} $S_{curr} \leftarrow S_{initial}$; $S_{best} \leftarrow S_{initial}$ for $t = 1, \dots, T$ do $s_t \leftarrow \text{ExtractState}(S_{curr})$ $a_t \leftarrow \text{Actor}\pi_\theta(s_t)$ $(d_k, r_h, params) \leftarrow \text{Decode}(a_t)$ $A \leftarrow \text{Destroy}(S_{curr}, d_k, params)$ $S_{new} \leftarrow \text{Repair}(S_{curr}, A, r_h, params)$ $F_{new} \leftarrow f(S_{new})$; $F_{curr} \leftarrow f(S_{curr})$ $R_t \leftarrow F_{curr} - F_{new}$ $V_t \leftarrow \text{Critic}v_\phi(s_t)$ $\delta_t \leftarrow R_t + \gamma \cdot v_\phi(\text{ExtractState}(S_{new})) - V_t$ If $F_{new} < F_{best}$ then $S_{best} \leftarrow S_{new}$; $F_{best} \leftarrow F_{new}$ end if if $F_{new} < F_{curr}$ then $S_{curr} \leftarrow S_{new}$ end if Update Critic: $\phi \leftarrow \phi + \alpha_2 \nabla_\phi [R + \gamma v_\phi(x') - v_\phi(x)]^2$ Update Actor: $\theta \leftarrow \theta + \alpha_1 \nabla_\theta \log \pi(a x) [R + \gamma v_\phi(x') - v_\phi(x)]$ end for return S_{best}

3.4. Reinforcement learning-guided adaptive large neighborhood search algorithm

Building upon the reinforcement learning-based environment and state design introduced in Section 3.3, this section develops a novel solution framework that integrates reinforcement learning with the ALNS algorithm.

The model was solved by using a RL-ALNS algorithm. The specific algorithm flow of the RL-ALNS is shown in Fig. 1. In this framework, $S_{initial}$ is the initial solution, S_{curr} is the current solution, S_{best} is the optimal solution, and S_{new} is the new solution obtained by the destruction and repair operation. The RL-ALNS algorithm selects multiple destruction and repair operators based on the quality of the solutions generated by the damage and repair operators. The pseudocode of the RL-ALNS algorithm is shown in Table 1. The RL-ALNS algorithm designed in this study is based on the standard ALNS framework and introduces the Actor-Critic module from reinforcement learning to dynamically optimize the operator selection strategy. This enhances the adaptability and convergence efficiency of the search.

For multiple distribution centers distributed across a certain area, initial paths are first planned based on static customer planning. When the real-time service time window opens, the distribution center will continuously process new customer orders received. Considering the complexity of dynamic problems, dynamic problems are first simplified

into several static subproblems. In this paper, the distributed time window is selected as multiple time periods. Each time period requires nodes to collect dynamic information. For new customers within a time segment, the entire system provides an initial path for serving customers, including uncompleted customers and new customers. As shown in Fig. 1, the distribution strategy is divided into three types.

If the current cargo capacity of the vehicle is insufficient to complete the subsequent customer delivery tasks, the system first extracts the current state vector s_t . The Actor module automatically selects the perturbation operator and repair operator based on the policy $\pi(a_t|s_t)$ to perturb the current solution and generate a new path solution S_{new} . If the new solution satisfies all customer time window constraints and its path cost is lower than the original path, the new customer task is assigned to the vehicle, and the vehicle path is updated to execute the new delivery plan.

When replenished vehicles are still unable to meet customer service time window requirements, or when the service costs incurred by replenishment exceed the costs of dispatching new vehicles directly from the distribution center, the system will determine that the current vehicles are no longer capable of performing the remaining tasks. At this point, the Actor module will reassess the status and select new scheduling actions, with the distribution center arranging for other available vehicles to perform subsequent service tasks. This ensures that overall service quality is not compromised.

Throughout the process, the algorithm scores the optimization effect of the current path and calculates the reward signal based on the difference between the objective function values of the new and old solutions. The state value function is evaluated through the Critic module, thereby updating the parameters of the Actor and Critic. This mechanism effectively improves the dynamic adaptability of the operator selection strategy, making path optimization more intelligent and stable.

The destruction operator makes partial removal of the customer points in the current solution S , in order to arrange more reasonable routes for the removed customers in A . The deletion list A is used to store the deleted customer points. This paper selects three damage operators, and the general pseudocode of the three damage operators are as shown in Table 2.

The repair operator is to insert the customer points removed by the damage process into the current solution in a better way, etc. The customer set B stores all the customer points to be served. This paper selects three repair operators, and the general pseudocode of the operator is shown in Table 3.

4. Experiment

In this paper, the RL-ALNS optimization model for the dynamic distribution path of agricultural products under replenishment demand is designed. Its validity and effectiveness are verified through large-scale data, and example results are further analyzed.

Table 2
Pseudocode of the damage operators.

Removal Heuristic
Input: solution S Initial: request list $A = \text{null}$ for removal request list $B = \text{requests which have been arranged in } S$ while $ A < q$ do sort B by certain rules select the first sorted point in B $A = A \cup i$ remove i from B end while return S

4.1. Data sources

Since this study focuses on an extended version of the traditional vehicle routing problem, the LINGO series standard examples are still used for validation. Following the work, which utilizes LINGO software, this study modifies and extends those examples to better suit the dynamic vehicle routing context. To adapt to the specific requirements of this study, the customers in the examples are categorized into static and dynamic customers, and a 24-hour time window is applied to the data. For small-scale data, the same data used in the model testing is utilized, testing both small and large-scale examples. The minimum satisfaction value remains set at 0.7. The large-scale example consists of 52 points, including 2 distribution center points, 30 static customers, and 20 dynamic customers.

4.2. Algorithm parameter analysis

The RL-ALNS algorithm is characterized by its adaptive scoring mechanism, which flexibly selects high-performing operators. Theoretically, the performance of operators is closely tied to their scores. The stronger the operator's performance, the higher the score it receives. The overall effectiveness of the algorithm is significantly enhanced when an appropriate scoring mechanism is adopted. Therefore, this section analyzes the RL-ALNS algorithm's scoring parameters δ_1 , δ_2 , δ_3 , and ζ . By reviewing relevant literature, five sets of well-validated scoring mechanisms have been identified. The best among these is selected to provide a basis for subsequent case study results analysis. The five sets of scoring criteria are listed in Table 4. These five sets of scoring mechanisms were used to evaluate both the small-scale and large-scale cases of this study. To minimize the impact of errors on the solution results, each case was tested three times, with S-1 referring to the first test of the small-scale case, S-2 to the second test, and S-3 to the third test, M-1 refers to the first test of the large-scale case, M-2 to the second test, and M-3 to the third test. The results of the three tests are presented in Table 5.

Through the calculation example tests, it was found that the scores of several groups did not show significant differences when testing small-scale examples. However, when testing large-scale examples, the differences became more apparent. Among these, the score set [1, 0.4, 0.25, 0.25] performed better than the other groups and is more suitable for the research question of this paper. Therefore, this paper selects this set of scores for subsequent data analysis. The removal ratio parameter for the number of customers in this paper has been adjusted through testing, and the specific parameter settings of the algorithm are shown in Table 6.

4.3. Result analysis of dynamic distribution path optimization

The small-scale test is used to verify the accuracy and effectiveness of the algorithm in this paper. After running and debugging, the path planning diagram is obtained. In this paper, the distribution problem is mathematically described as a Mixed-Integer Nonlinear Programming (MINLP) model, which is then solved using the proposed RL-ALNS algorithm. The RL-ALNS algorithm achieves an optimal objective value of 525.278 with a total computation time of 38 s. In the experimental setting, the update frequency of the distribution route is set to once per hour. The delivery service starts at 8:00, and delivery personnel receive dynamic customer requests at 8:30 and 9:17, respectively. As shown in Fig. 2, the system includes 2 distribution centers (red circles), 2 distribution vehicles (not shown in the figure), 5 static customers (purple diamonds), and 2 dynamic customers (green triangles). After receiving the dynamic distribution request, the distribution route will change somewhat. The route of the system for the initial static customer planning is shown in Fig. 2a. After receiving the delivery request from the first dynamic customer at 8:30, the path of the delivery vehicle at 9:00 is replanned, as shown in Fig. 2b. At 9:17, the second dynamic request

Table 3
Pseudocode of the repair operators.

Removal Heuristic
Input: solution S Initial: request list B for insertion while $ B > 0$ do select a request i from B by certain rules if insert into S is feasible Insert i Into S end if remove from B end while return S

Table 4
Scoring mechanism table.

Parameter	Numerical value
$[\delta_1, \delta_2, \delta_3, \zeta]$	[40, 30, 20, 0.25]
$[\delta_1, \delta_2, \delta_3, \zeta]$	[33, 13, 9, 0.35]
$[\delta_1, \delta_2, \delta_3, \zeta]$	[1, 0.4, 0.25, 0.25]
$[\delta_1, \delta_2, \delta_3, \zeta]$	[100, 20, 10, 0.5]
$[\delta_1, \delta_2, \delta_3, \zeta]$	[20, 12, 8, 0.8]

from the customer was received. The delivery path was subsequently updated and finalized at 10:00, as shown in Fig. 2c.

The final path is shown as ②—5—4—2—1—6—7—①(supplement)—3—②.

To evaluate the performance of an algorithm, three key factors are typically considered including the speed of solution, the accuracy of the solution, and the stability of the solution. As shown in Table 7, compared to the traditional LINGO solver, the RL-ALNS algorithm proposed in this paper demonstrates significant advantages in multiple key sub-metrics while maintaining the total objective cost unchanged. First, RL-ALNS achieves slight reductions in transportation costs, carbon emissions costs, and cargo damage costs, decreasing from 310.4 RMB, 10.3 RMB, and 4.23 RMB to 308.5 RMB, 9.8 RMB, and 4.07 RMB, respectively. This indicates that the algorithm exhibits higher performance and adaptability in path optimization and scheduling accuracy. Although there was a slight increase in time window penalty costs, this was primarily due to the algorithm allowing local compromises in its global search to achieve a more optimal overall solution. Meanwhile, the customer satisfaction metric improved from 0.96 to 0.97, further validating the algorithm's optimization effects on service levels. More notably, RL-ALNS demonstrated significant efficiency advantages in runtime, with a solution time of just 7.04 s. Compared to LINGO's 28 s, this represents a substantial reduction, validating the effectiveness of reinforcement learning mechanisms in guiding search strategies. In summary, the RL-ALNS algorithm not only offers a more reasonable cost structure but also demonstrates stronger overall competitiveness in terms of computational efficiency and service quality.

In the large-scale experiment analysis, the operation scale is 52 nodes, including 2 distribution centers, 30 static customers, 20 dynamic customers and 4 dynamic update nodes. To obtain better update results, the number of iterations for this case is set to 200. Under the same running environment, the initial distribution path, the iteration results of the initial path algorithm, and the final distribution path are shown in

Table 5
Test results of operator scoring parameters.

Example	S-1	S-2	S-3	M-1	M-2	M-3	average value
[40, 30, 20, 0.25]	707.03	707.03	707.03	2043.50	1943.26	1981.54	1348.23
[33, 13, 9, 0.35]	707.03	707.03	707.03	1968.40	1946.58	2055.83	1348.65
[1, 0.4, 0.25, 0.25]	707.01	707.03	707.01	1849.35	1821.05	1838.07	1271.58
[100, 20, 10, 0.5]	707.01	707.01	707.03	1950.27	2185.50	1960.58	1369.67
[20, 12, 8, 0.8]	707.01	707.03	707.03	1962.16	2049.92	1973.16	1351.05

Fig. 3.

① and ② represent two distribution centers, ①(supplement) and ②(supplement) represent vehicles returning to the distribution centers for replenishment. Arabic numerals indicate the customer node numbers. The results of the distribution of the calculated examples are analyzed according to the order of the dynamic update cycle. Based on the above model analysis.

Table 8 shows the initial path of the dynamic distribution of agricultural products and the updated path at 9:00. Between 8:00 and 9:00, six dynamic customers submitted distribution requests. Therefore, the path optimization included six new distribution tasks numbered 31–36. The algorithm-optimized path, including the added replenishment route, is shown as the “Final path” in Table 8.

Table 9 shows the final path optimization results. The path is relatively simple, all distribution tasks are fully completed, and all vehicles return to the distribution center.

It can be seen from Table 10, the lowest total cost of the scale calculation is 2246.641 RMB, and the shortest running time is 363.24 s. It can be observed that as the node scale, update frequency, and iteration number increase, the search space of the update algorithm expands, leading to higher computational complexity. To improve scalability in large-scale systems, future work will integrate parallel computing strategies and heuristic pre-processing to reduce the dimensionality of the solution space.

According to the results of the medium scale calculation shown in Table 10, the average value of the 5 runs is 2272.1834 RMB, the average running time is 375.146 s, and the average value of customer satisfaction across the five runs is 0.9409. Among them, the optimal solution is 2246.641 RMB, the shortest running time is 363.24 s, the average deviation of the running results is 1.136 %, and the largest deviation is 1.80 %. These results fall within a reasonable range, further indicating that the algorithm is excellent in stability.

Based on the analysis of this example, the vehicle's distribution routes and supply conditions are continuously adjusted in response to dynamic demand and update cycles. The fourth update, in particular, provides better service quality compared to the previous ones. The distribution center left the distribution work again and arranged the correct distribution route after the update at 14:00. After customer service is completed, four vehicles return to their respective distribution centers. The subsequent dynamic customer demand is primarily served by other vehicles. This approach uses fewer vehicles to complete the dynamic distribution task, thereby saving more starting costs. Meanwhile, as the number of updates increases, the distribution routes tend to become more stable. The changes in vehicle service assignments gradually decrease, which also reflects the optimization process of the

Table 6
Parameter setting.

Parameter	Definition	Short-cut process
q_1	Remove the percentage of customers	1 %
q_2		15 %
q_3		30 %
δ_1	Operator scoring parameters	1
δ_2		0.4
δ_3		0.25
ζ	Adjust the operator weight parameter	0.25

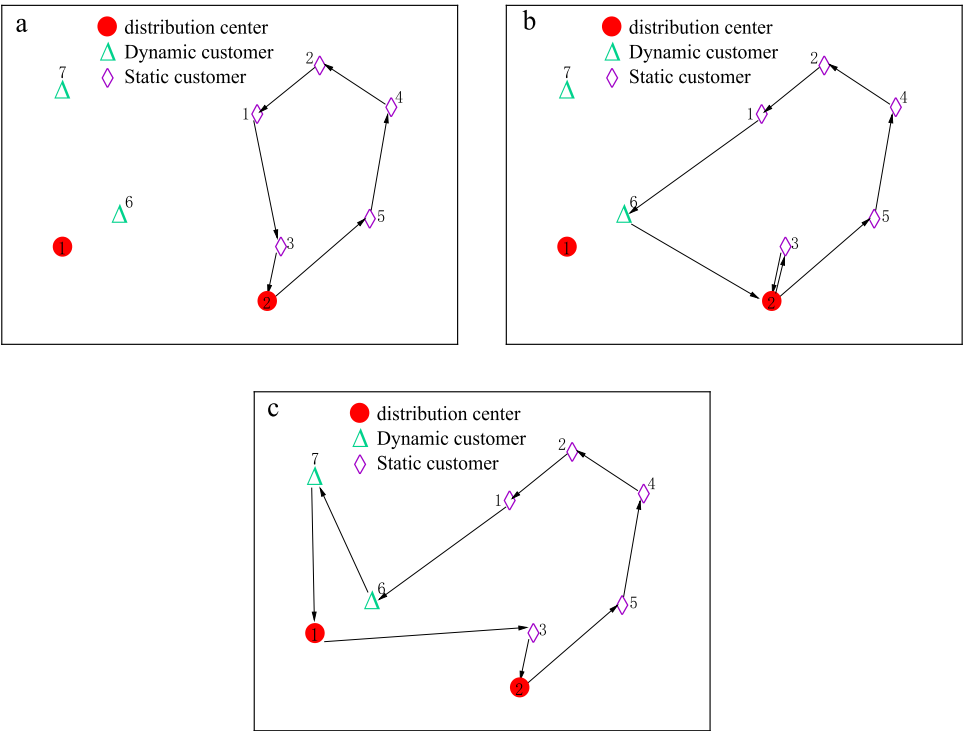


Fig. 2. Schematic diagram of the pathway planning. a. Initial route planning diagram, b. Distribution path diagram after the first update, c. Distribution path diagram after the second update.

Table 7
Comparison between the RL-ALNS algorithm in this paper and LINGO results.

Target	LINGO	RL-ALNS
Objective function (RMB)	525.3	525.3
Fixed cost (RMB)	200	200
Transport cost (RMB)	310.4	308.5
Carbon emission cost (RMB)	10.3	9.8
Cargo loss cost (RMB)	4.23	4.07
Time window penalty cost (RMB)	0.25	0.48
Customer satisfaction	0.96	0.97
performance period (s)	28	7.04

algorithm.

4.4. Example analysis

To further demonstrate the effectiveness of the proposed RL-ALNS, an additional case study is conducted. The customer data used in this case are taken from a selected company, referred to as X Agricultural Products Company (X Company), covers a wide range of agricultural

products and maintains a well-developed logistics distribution system. It provides direct delivery services of agricultural products to large, medium, and small fresh supermarkets across the city. In this experiment, the distribution data of X Company on a given day are adopted. On that day, a large distribution center served a total of 22 customers, including 20 static customers and 2 new dynamic customers. The rated load capacity of the transport vehicles used by the company is 1495 kg. After 100 iterations under the same computational environment as the

Table 8
Specific distribution route.

Initial path	①-19-8-11-12-20-9-① ②-13-27-22-23-24-5-26-25-30-② ②-4-10-21-2-28-1-29-② ①-18-17-1516-3-14-7-6-①
Final path	①-19-8-11-31-33-12-32-2-34-②(supplement)-29-1-28-49-20-50-37-48-47-①(supplement)-46-① ②-13-27-22-23-24-5-26-25-②(supplement)-42-43-44-39-45-38-41-② ②-4-10-21-36-35-30-② ①-18-17-15-16-3-14-7-6-①(supplement)-9-①

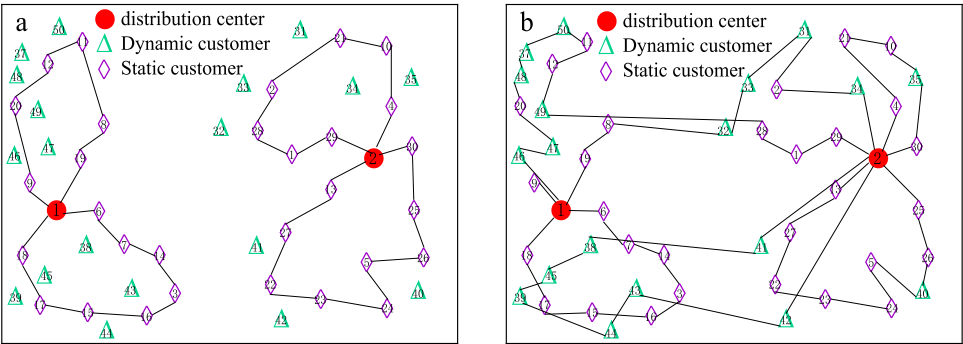


Fig. 3. Schematic diagram of the pathway planning. a. Initial route planning diagram, b. Final distribution path diagram.

Table 9
Distribution path after the final update.

Vehicle	Path
1	①–19–8–11–31–33–12–32–2–34–②(supplement)–29–1–28–49–20–50–37–48–47–①(supplement)–46–①
2	②–13–27–22–23–24–5–40–26–25–②(supplement)–42–43–44–39–45–38–41–②
3	②–4–10–21–36–35–30–②
4	①–18–17–15–16–3–14–7–6–①(supplement)–9–①

Table 10
Run the results 5 times.

Order number	Results (RMB)	Run time (s)	Discrepancy (%)	Customer Satisfaction
1	2283.442	374.31	1.61 %	0.9412
2	2246.641	363.24	-	0.9407
3	2287.325	368.92	1.80 %	0.9401
4	2258.082	373.86	0.50 %	0.9398
5	2285.427	395.40	1.77 %	0.9426
Average	2272.1834	375.146	1.136 %	0.9409

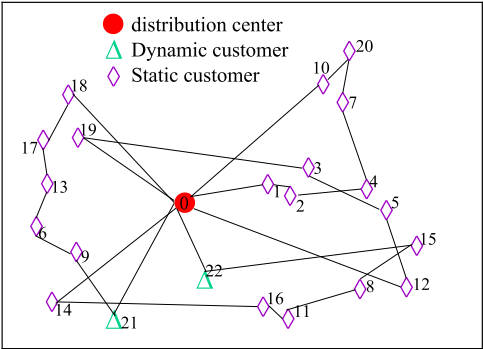


Fig. 4. Example customer final distribution path diagram.

previous example, the optimized distribution route is obtained, as shown in Fig. 4. For comparison, the distribution route reported in X Company is shown in Fig. 5.

As the original calculation cost is different from that reported in X Company, the comparison in Table 11 is restricted to the fixed cost and the time window penalty cost. As shown in Table 11, the proposed RL-ALNS achieves a total cost of 400 RMB, which is 100 RMB lower than that of X Company (500 RMB). Although the time window penalty cost in this study is 53.2143 RMB (higher than the 0 RMB in the reference), the total cost of RL-ALNS is 453.2143 RMB (400 + 53.2143), still lower than the 500 RMB in the reference. This corresponds to an overall reduction of approximately 9.33 %. This reduction is mainly due to the ability of the proposed algorithm to allow vehicles to return to distribution centers for replenishment, thereby saving departure and vehicle usage costs. Moreover, the customer satisfaction remains high at 0.9637, indicating improved performance even with fewer vehicles (4 vs. 5). The reason for the result is that this paper allows the vehicle, in the case of insufficient goods, to return to the distribution center for replenishment. This saves the departure cost. Meanwhile, the vehicle takes additional time for replenishment, which increases the time window penalty cost. However, even considering these two costs, the proposal still reduces the

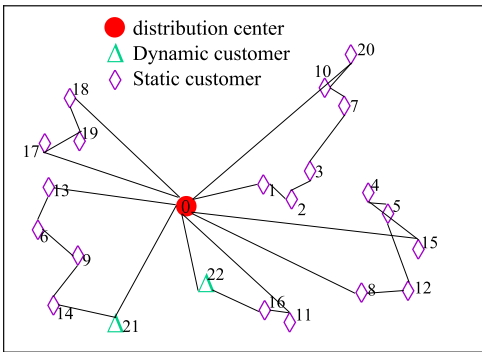


Fig. 5. Distribution path diagram of X Company.

Table 11
Comparison of this proposed method with the state-of-art method.

Target	RL-ALNS	Reference
Cost (RMB)	400	500
Time window penalty cost (RMB)	53.2143	0
Customer satisfaction	0.9637	-

overall cost by approximately 9.33 %. At the same time, the RL-ALNS maintains a high customer satisfaction rate. Therefore, it can be seen that the RL-ALNS proposed in this paper demonstrates good practicality, effectiveness, and superiority.

4.5. Comparative analysis

In order to reflect the superiority of the multi-distribution center replenishment logistics distribution scheme considering the dynamic demand, the RL-ALNS is compared with the multi-distribution center scheme that does not allow dynamic demand during replenishment. Because vehicles are not allowed to replenish along the route, a vehicle with insufficient capacity cannot serve the customer. In such cases, the demand can only be met either by another vehicle with sufficient remaining capacity or by dispatching a new vehicle from the distribution center under all constraints. A small-scale example is selected as comparison data to obtain more representative results, and the results of the scheme without along-route replenishment are shown in Fig. 6.

By Table 12, under the same customer scale, the service cost without considering the en-route replenishment strategy is higher than that with the en-route replenishment strategy. Without considering en-route replenishment, the final distribution cost is 2920.1183 RMB, which is about 29.99 % higher than the 2246.3215 RMB incurred when considering en-route replenishment. At the same time, the customer satisfaction of the delivery mode without en-route replenishment is about 7.84 % lower than that with en-route replenishment. In terms of various costs, the distribution scheme without en-route replenishment shows a transportation cost about 5.03 % higher than that with en-route replenishment. The main reason for the high cost without en-route replenishment is the increase in the number of vehicles used. Moreover, although more vehicles are dispatched from the distribution center to serve customers, customer satisfaction is not improved and is actually lower than that under the en-route replenishment delivery plan.

The above results show that the RL-ALNS considering the replenishment strategy along the way has certain advantages. The proposed RL-ALNS can reduce the number of vehicles and reduce the distribution

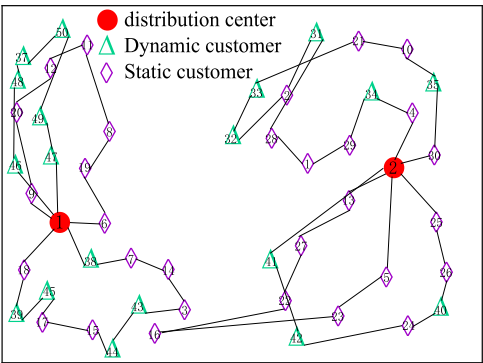


Fig. 6. Optimized distribution route without en-route replenishment.

Table 12
Results of scale example runs.

Target	No replenishment was considered	Consider replenishment
Total cost (RMB)	2920.1183	2246.3215
Constant cost (RMB)	1400	800
Transportation cost (RMB)	1460.2234	1390.3451
Carbon emission cost (RMB)	48.6331	46.3578
Cost of goods loss (RMB)	1.6522	1.6074
Time window penalty cost (RMB)	9.2456	8.9435
Customer satisfaction	0.8613	0.9346

cost of enterprises, while maintaining customer satisfaction. For the dynamic demand distribution model, the dynamic update time determines the timeliness of distribution. The length of dynamic update time greatly affected the vehicle of dynamic customer service time and path arrangement, which in turn changes the final distribution cost. To make the results more representative, this paper selects large-scale examples with more update examples and sets four update time. The analysis of these four update times, along with the corresponding running results as shown in Table 13. The change trend is shown in Figs. 7 and 8.

The number of dynamic customers has a great impact on the dynamic demand distribution. This section analyzes the impact of the number of dynamic customers on the distribution path, using the dynamic ratio to represent the proportion of dynamic customers. The method of dynamic attitude is to express the number of dynamic customers divided by the total number of dynamic customers, the more the number of dynamic customers, the greater the dynamic attitude. In this paper, 20 dynamic customers and 30 static customers are considered, resulting in a dynamic ratio of 0.4. By changing the number of dynamic customers in the medium-scale case, several instances with different dynamic ratios are solved. The operation results are shown in Table 14, and the change trend is shown in Figs. 9 and 10.

From the results of the above analysis, as the number of dynamic customers increases, the total distribution costs of the path also increase. This is because the increase in dynamic customers leads to more frequent path replanning. In addition, the growing number of dynamic

Table 13
Dynamically update the time sensitivity analysis results.

Refresh time (min)	Cost (RMB)	Customer satisfaction	Computation period (s)
30	4374.2541	0.9544	2013.44
60	4056.2123	0.9431	1732.31
90	3851.6471	0.9233	1641.54
120	3752.6632	0.9044	1543.59
150	3613.5316	0.8933	1471.33
180	3528.3521	0.8840	1413.49

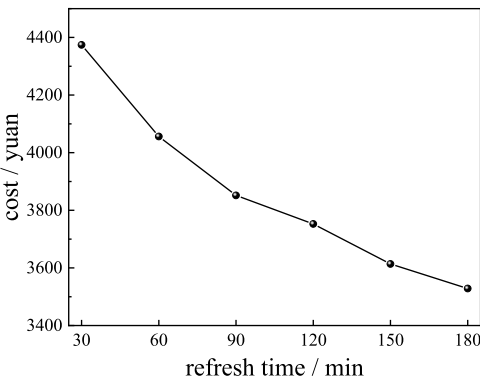


Fig. 7. The trend of the cost with the dynamic update time.

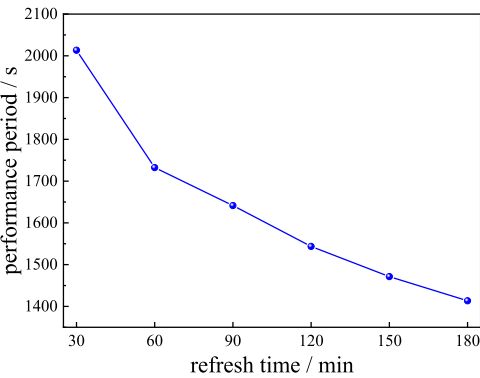


Fig. 8. The trend of running time with dynamic update time.

Table 14
Sensitivity analysis of the number of dynamic customers.

Move attitude	Cost (RMB)	Performance period (s)
0	1742.5483	284.74
0.2	1959.2954	329.72
0.4	2246.8821	363.24
0.6	2452.3561	412.51
0.8	2634.2431	441.22
1.0	2877.5813	473.46

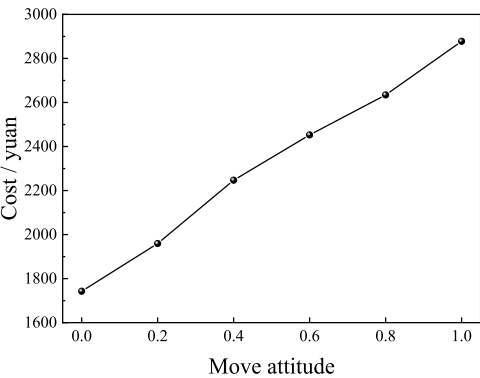


Fig. 9. The trend of the cost with the move attitude.

customers increases the frequency of vehicle replenishments or new shipments. Therefore, the total cost of distribution inevitably rises. At the same time, when the number of dynamic customers increases, the running time of the algorithm also increases. This is because, when there

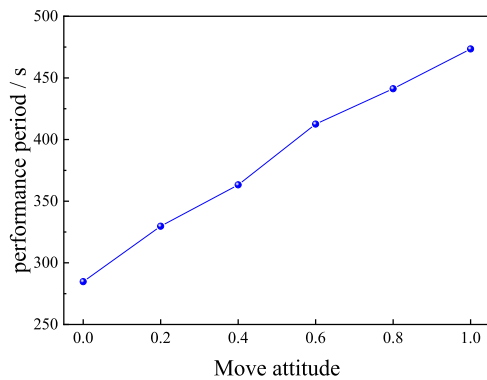


Fig. 10. The trend of running time with the move attitude.

are fewer dynamic customers, most of the customers in the path are static. Better path schemes can be found before dynamic updates occur. Thus, when only a small number of dynamic customers are added, the path changes relatively little. However, when the number of dynamic customers is very large, the path is updated more frequently, and it takes a longer runtime to complete path replanning.

5. Conclusion

Based on the extended study of multi-node logistics distribution incorporating dynamic demand and en-route replenishment. A cold chain logistics distribution model for multi-distribution centers is established, addressing both dynamic customer demand and en-route replenishment. The detailed construction process of the RL-ALNS algorithm is given. By comparing and selecting an appropriate adaptive scoring mechanism, this paper designs a mechanism that flexibly determines the proportion of removed customers, thereby enhancing the problem-solving ability of the proposed algorithm. The accuracy, stability, and effectiveness of the RL-ALNS algorithm are verified through computational experiments. In the comparative analysis, two distribution schemes were evaluated, one considering the replenishment strategy along the way and the other without it. The results show that considering the replenishment strategy reduces the total cost by 29.99 %, decreases the number of departures by 3, and increases customer satisfaction by 7.84 %. These findings demonstrate that the replenishment strategy has strong applicability under dynamic demand conditions.

CRediT authorship contribution statement

Delong Zhu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Qianyang Liu:** Visualization, Validation, Software, Resources, Project administration, Funding acquisition, Data curation. **Dafa Zuo:** Writing – original draft, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis. **Lujiang Yin:** Visualization, Validation, Software, Resources, Methodology, Investigation. **Baolin Li:** Writing – original draft, Visualization, Software, Resources, Project administration, Methodology. **Jiang Hu:** Writing – original draft, Visualization, Validation, Supervision, Software, Funding acquisition, Formal analysis. **Caozheng Yan:** Writing – original draft, Validation, Supervision, Data curation. **Bo He:** Writing – original draft, Visualization, Software, Resources, Project administration. **Changchun Xue:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

There is no conflicts of interest.

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